

Muhammad Bilal Hussain

bilalhussain42201@gmail.com | [+923352009245](tel:+923352009245) | [Portfolio](#) | [LinkedIn](#) | [GitHub](#) | [HuggingFace](#)

EDUCATION

Bachelor of Science in Software Engineering,
Sindh Madressatul Islam University

Feb 2027 – Present
Karachi, Pakistan

EXPERIENCE

Front-end AI Engineer Intern, FlyRank AI (Remote)

Jun 2026 – Sep 2026

- Developed AI-powered frontend applications using React, Next.js, and TypeScript, integrating AI capabilities and APIs.
- Built responsive and accessible user interfaces while improving performance, testing, documentation, and deployment workflows.

Remote

HACKATHONS

DYLP Vibe Coding Hackathon 2026 [🔗](#), AI & Full-Stack Engineer

- Achievement:** Top 30 Innovative Projects
- Project:** Developed **Gulzareen**, an AI-powered crop-residue classification platform using Computer Vision and FastAPI to identify agricultural waste and recommend reuse opportunities.

AMD Developer Hackathon: ACT II [🔗](#), AI Engineer

- Achievement:** Track 2 – Video Captioning
- Project:** Developed an AI-powered video captioning pipeline using Vision Language Models, FastAPI, and FFmpeg to automate multi-style caption generation.

NativeBuilder: Build Without Limits [🔗](#), AI Developer

- Achievement:** Participant (AI Factory)
- Project:** Developed an innovative AI-powered solution during the hackathon.

PROJECTS

Gulzareen – AI Crop Residue Classification Platform [🔗](#)

- Developed an AI-powered crop residue classification platform using Computer Vision, ONNX Runtime, and FastAPI, delivering real-time image classification results.
- Integrated AI insights for market value estimation, reuse opportunities, contamination detection, and CO₂ impact analysis, with backend documentation, exception handling, and deployment workflows.

AMD Video Captioning AI [🔗](#)

- Built an AI-powered video captioning pipeline using Vision Language Models, FastAPI, and FFmpeg.
- Generated multiple caption styles including formal, humorous, and creative outputs from video inputs.
- Integrated LLM-based evaluation workflows for caption quality assessment.

RAG Document QA System (GROQ + FAISS) [🔗](#)

- Built an end-to-end RAG document QA system using Streamlit, SentenceTransformers, and FAISS for PDF processing, embeddings, and fast similarity-based retrieval.
- Integrated Groq API (Llama 3.1) to generate context-aware answers from retrieved document content and deployed the application as a functional web app.

RedStore – E-Commerce Frontend Platform [🔗](#), Frontend Project

- Built a responsive multi-page e-commerce website using HTML, CSS, and JavaScript.
- Designed product pages, cart system, and UI components.
- Ensured mobile, tablet, and desktop compatibility; deployed on Vercel.

SmartHire – AI Recruitment & ATS Platform [🔗](#), AI Engineer & Full-Stack Developer

- Built an AI-powered recruitment platform using React, FastAPI, PostgreSQL, and Gemini for resume analysis and candidate matching.
- Added human-in-the-loop scoring, RBAC, multi-tenant ATS workflows, and explainable AI recommendations.

Civic Voice – AI-Powered Civic Complaint Generator [🔗](#), AI & Full-Stack Developer

- Built a Python/Streamlit AI application using Google Gemini to transform informal civic issues into structured, professional complaint letters.
- Implemented smart priority detection, department routing, TXT/DOCX/PDF export, email submission, and QR-code support with template fallback for AI failures.

Flight Assistant – AI-Powered Multi-Agent Flight Booking Assistant [🔗](#),

Full-Stack & Agentic AI Engineer

- Built a full-stack AI flight assistant using Next.js, FastAPI, Python, OpenAI Agents SDK, Gemini, and PostgreSQL/Neon with multi-agent orchestration and tool guardrails.
- Implemented AI flight search, multi-turn sessions, JWT authentication, booking & passenger management, Alembic migrations, and Vercel production deployment.

CERTIFICATES

- FlyRank AI – Front-End AI Engineering Internship Certificate [🔗](#)
- FlyRank AI – AI Fluency Track Certificate [🔗](#)
- CS50x Puzzle Day 2026 – Harvard University [🔗](#)
- Stanford Code in Place – Programming Methodologies [🔗](#)
- Postman API Fundamentals Student Expert [🔗](#)
- Legacy Responsive Web Design V8 (freeCodeCamp) [🔗](#)
- Python For Data Science – Udemy [🔗](#)

SKILLS

AI & Machine Learning: Generative AI, Agentic AI, RAG, Vector Search (FAISS), Prompt Engineering, LLM Integration, Computer Vision

Programming Languages: Python, JavaScript (ES6+), TypeScript

Frontend: HTML5, CSS3, React, Next.js

Frameworks & Backend: FastAPI, LangChain, LangGraph, SQLAlchemy, REST APIs, JWT Authentication

Tools & Platforms: Git, GitHub, Docker, Postman, Streamlit, Gradio, Vercel, Netlify